

When gold disconnects: the size bound, Dirichlet, and the sinks

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Abstract

The hire graph of the 3-free door is built from two elementary inputs. The first is a *size bound*: if an odd prime q divides the door of a prime p , then $p \geq 2q - 1$, so every gold arc runs from a larger prime to a smaller one. The second is *Dirichlet's theorem*, applied to two residue classes, which makes every owner set infinite. From the size bound alone, every gold component of a finite window contains a 2-power door (a Mersenne or Fermat prime), so connectivity of gold is exactly the question of whether the 2-power doors present are joined. From Dirichlet, any two of them are joined in some finite window; the size bound shows how late that can be. Consequently gold is disconnected for infinitely many windows if and only if there are infinitely many Mersenne or Fermat primes, and each 2-power door $M \geq 7$ forces a whole *octave* $[\omega(M), 2\omega(M) - 2]$ of disconnected windows, not merely one. For M_{31} the picture is exact: assuming the machine-checked connectivity at $X^* = 92\,274\,421$ of the hire-graph note, gold is disconnected for $X^* \leq X \leq 2M_{61} - 2$ precisely when $98\,784\,247\,763 \leq X \leq 627\,065\,224\,920$.

1 Two inputs

Everything in this note is a consequence of two facts about the door $m_0(p) = p + \chi_3(p)$.

The size bound. The door of p is an even number at most $p + 1$. An odd prime q dividing it therefore divides half of it, so $q \leq (p + 1)/2 < p$. Read from the other side: an owner of q is at least $2q - 1$. This single inequality (Lemma 3.1) is used in four ways: it makes gold acyclic, it caps the primes in a window by $(X + 1)/2$, it gives *door closure* (Corollary 3.3), and it makes every downward gold path terminate. The last use is Theorem 3.5: components of gold are governed by the primes with power-of-two doors.

Dirichlet. Whether $Q \mid m_0(p)$ depends only on p modulo $6Q$, and it selects exactly two invertible classes (Lemma 4.1). Dirichlet's theorem then gives infinitely many owners of any Q coprime to 6, and applying it twice (a common owner, then a hirer of that owner) joins any two primes in some finite window (Theorem 4.3). Dirichlet is qualitative. The size bound is what turns “eventually” into quantitative lower bounds, and Linnik's theorem is the only source of upper bounds.

Results.

- *Sink reduction* (Theorem 3.5): for every window $X \geq 11$, each gold component contains a 2-power door, and gold is connected iff the 2-power doors present are pairwise joined.
- *Isolation octaves* (Theorem 5.1): if $M \geq 7$ is a 2-power door with first owner $\omega(M)$, then gold is disconnected on all of $[\omega(M), 2\omega(M) - 2]$.
- *Equivalence* (Theorem 6.2): gold is disconnected for infinitely many X iff there are infinitely many Mersenne or Fermat primes.

- M_{31} , *exactly* (Theorem 7.1): the complete disconnected stretch, its end point $X_1 = 627\,065\,224\,921$, and the reason the bridge owner v of the previous version of this note never bridges.

The infinitude of Mersenne primes is open, so item (1) of Theorem 6.2 is not settled unconditionally.

2 Windows, owners and sinks

Notation follows the hire-graph note [1]. For a prime $p \geq 5$ let $\chi_3(p) = +1$ if $p \equiv 1 \pmod{3}$ and -1 if $p \equiv 2 \pmod{3}$, and write $\chi_3(n)$ likewise for any n coprime to 3.

Lemma 2.1 (Door basics). *Let $p \geq 5$ be prime and $m = m_0(p)$.*

(a) $m \in \{p-1, p+1\}$, m is even, $3 \nmid m$, and $m \geq 4$.

(b) (Inversion) $p = m + \chi_3(m)$.

Proof. (a) If $p \equiv 1 \pmod{3}$ then $m = p+1 \equiv 2 \pmod{3}$; if $p \equiv 2 \pmod{3}$ then $m = p-1 \equiv 1 \pmod{3}$. Evenness and $m \geq 4$ are clear. (b) If $p \equiv 1 \pmod{3}$ then $m = p+1 \equiv 2$, so $\chi_3(m) = -1$ and $m + \chi_3(m) = p$. If $p \equiv 2 \pmod{3}$ then $m = p-1 \equiv 1$, so $\chi_3(m) = +1$ and $m + \chi_3(m) = p$. $\square$

We say p *owns* an odd prime $q \neq 3$ if $q \mid m_0(p)$; such q is automatically ≥ 5 . For $X \geq 5$ put $P_X = \{p \text{ prime} : 5 \leq p \leq X\}$ and

$$S_X = \{2\} \cup \{q \text{ odd prime} : q \mid m_0(p) \text{ for some } p \in P_X\}.$$

The *gold graph* G_X has vertex set $S_X \setminus \{2\}$ and a directed arc $p \rightarrow q$ whenever $q \mid m_0(p)$; the undirected gold graph forgets orientation. Connectivity always refers to the undirected gold graph. (The hire graph proper is the cone over this graph with apex 2, which is why finite windows are connected whether or not gold is; see [1].)

Let $\omega(q)$ denote the least owner of q , the *first owner*, with $\omega(q) = \infty$ if none exists (Corollary 4.2 shows this never happens). By definition

$$q \in S_X \iff \omega(q) \leq X. \tag{1}$$

So $\omega(q)$ is the *hire time* of q : the first window in which q is a vertex. In this notation the “first owner r_0 of M ” of the earlier version is $\omega(M)$; we keep r_0 for $\omega(M_{31})$.

Since arcs between two vertices of S_X are determined by the doors alone, we have monotonicity:

$$X \leq X' \implies S_X \subseteq S_{X'} \text{ and } G_X \text{ is an induced subgraph of } G_{X'}. \tag{2}$$

In particular two vertices joined in G_X stay joined in every larger window. Connectivity of the *whole* graph is not monotone: a new vertex can enter with no arcs.

A prime $M \neq 3$ is a *2-power door* if $m_0(M) = 2^k$. Write $\mathcal{M}$ for the set of these. By the hire-graph note (Lemma 10 there), $\mathcal{M}$ consists of the Mersenne primes $2^p - 1$ (p odd) and the Fermat primes $2^{2^n} + 1$, $n \geq 1$:

$$\mathcal{M} = \{5, 7, 17, 31, 127, 257, 8191, 65537, 131071, 524287, 2^{31} - 1, \dots\}.$$

A *sink* of G_X is a vertex with no outgoing arc. Note that $5 \in \mathcal{M}$, and that $\omega(5) = 11$: the doors of 5, 7 are 4, 8, and $m_0(11) = 10$.

3 The size bound

Lemma 3.1 (Size bound). *Let p, q be odd primes, $p \geq 5$, $q \neq 3$, with $q \mid m_0(p)$. Then*

$$2q \leq m_0(p) \leq p+1, \quad \text{hence} \quad q \leq \frac{m_0(p)}{2} \leq \frac{p+1}{2} < p, \quad p \geq 2q-1.$$

Proof. By Lemma 2.1, $m_0(p)$ is even and at most $p+1$. Since q is odd and divides an even number, $2q \mid m_0(p)$, so $m_0(p) \geq 2q$. Combine, and note $(p+1)/2 < p$ for $p > 1$. $\square$

The proof uses only that the door is even and at most $p+1$. Note the inequality is $q \leq (p+1)/2$, not $q < (p+1)/2$: equality occurs (for instance $p = 13$, $q = 7$), and the weaker consequence $q < p$ is all that is needed below.

Remark 3.2 (Sharpness). Equality $q = m_0(p)/2$ means $m_0(p) = 2q$. By Lemma 2.1(b) the only prime that can have door $2q$ is $p = 2q + \chi_3(2q) = 2q - \chi_3(q)$, that is $2q - 1$ if $q \equiv 1 \pmod{3}$ and $2q + 1$ if $q \equiv 2 \pmod{3}$. So $\omega(q) \geq 2q - \chi_3(q)$, with equality exactly when that number is prime. The next possible owners have door $4q$ and are $4q \pm 1$. This is the Sophie Germain pattern of the hire-graph note (Lemma 4 there) seen as the extremal case.

Corollary 3.3 (Consequences of the size bound). (a) *Acyclicity. Every gold arc $p \rightarrow q$ has $q < p$; the graph is acyclic and no two vertices are joined by arcs in both directions.*

(b) *Ceiling and floor. If $q \in S_X \setminus \{2\}$ then $q \leq (X+1)/2$. Equivalently $\omega(q) \geq 2q - 1$ for every odd prime $q \neq 3$.*

(c) *Door closure. If $q \in S_X \setminus \{2\}$ and r is an odd prime dividing $m_0(q)$, then $r \in S_X$.*

(d) *Chains of owners. If w owns q then $\omega(w) \geq 2w - 1 \geq 2\omega(q) - 1$.*

Proof. (a) is the inequality $q < p$. (b) Let $q \in S_X$, so $q \mid m_0(p)$ for some $p \leq X$; then $q \leq (p+1)/2 \leq (X+1)/2$. The equivalent form is (1). (c) Such q is a prime ≥ 5 with $q \leq (X+1)/2 \leq X$, so $q \in P_X$; every odd prime factor of its door is then in S_X by definition. (d) $\omega(w) \geq 2w - 1$ by (b), and $w \geq \omega(q)$ because w is an owner of q . $\square$

Parts (b) and (d) are the quantitative half of the note: a vertex cannot be hired before twice its own size, and a vertex hired through an owner w is hired no earlier than $2w - 1$, so delays double at every step of an ownership chain.

Proposition 3.4 (Descent). *Let $v \in S_X \setminus \{2\}$. There is a strictly decreasing gold path $v = v_0 \rightarrow v_1 \rightarrow \dots \rightarrow v_n$ in G_X ending at a vertex v_n with $m_0(v_n)$ a power of 2; that is, $v_n \in \mathcal{M}$.*

Proof. If $m_0(v_i)$ has an odd prime factor r , then $r \in S_X$ by Corollary 3.3(c), so $v_i \rightarrow r$ is an arc of G_X , and $r < v_i$ by (a). A strictly decreasing sequence of primes terminates; at the terminal vertex no odd prime divides the door. $\square$

Theorem 3.5 (Sink reduction). *Let $X \geq 11$ and put $D_X = \{M \in \mathcal{M} : \omega(M) \leq X\}$.*

(i) *D_X is the set of sinks of G_X , and $5 \in D_X$.*

(ii) *Every vertex of G_X is joined to some element of D_X by a decreasing gold path.*

(iii) *Every connected component of G_X contains an element of D_X . Consequently the number of components of G_X equals the number of classes of D_X under connectivity, and G_X is connected if and only if the elements of D_X lie in one component.*

Proof. (i) By (1), $M \in S_X$ iff $\omega(M) \leq X$. A vertex q has no outgoing arc iff no odd prime dividing $m_0(q)$ lies in S_X , iff (by closure) $m_0(q)$ has no odd prime factor, iff $q \in \mathcal{M}$. Since $\omega(5) = 11 \leq X$, $5 \in D_X$. (ii) is Proposition 3.4. (iii) follows from (ii); the count is then immediate. $\square$

The theorem says: the only obstruction to connectivity is between sinks. Generic vertices are always attached, through their descent, to a sink.

Remark 3.6 (Check). I replayed every window $X \leq 10^7$ (all 664 577 primes) with union-find and compared the component count of G_X with the number of classes of D_X : they agree at every window (`hire_checks.py`, part A). This is a test of Theorem 3.5, whose proof above is complete, and it also reproduces the component counts of the hire-graph note at $X = 30, 50, 100, 200, 10^4, 5 \cdot 10^5, 10^7$, namely 2, 2, 4, 4, 2, 3, 3.

4 Dirichlet

Fix an odd integer $Q \geq 3$ with $3 \nmid Q$, and put $s = \chi_3(Q)$, so $Q \equiv s \pmod{3}$.

Lemma 4.1 (Two classes). *For a prime $p \geq 5$,*

$$Q \mid m_0(p) \iff p \equiv 2Q - s \text{ or } p \equiv 4Q + s \pmod{6Q}.$$

Moreover the owners of Q are exactly the primes among the numbers

$$p_j(Q) = 2jQ - \chi_3(j)s, \quad j \geq 1, 3 \nmid j, \quad (3)$$

where $m_0(p_j) = 2jQ$; the first class corresponds to $j \equiv 1 \pmod{3}$ and the second to $j \equiv 2 \pmod{3}$.

Proof. Suppose $Q \mid m_0(p) =: m$. Then $2Q \mid m$ (m even, Q odd), say $m = 2jQ$, and $3 \nmid j$ because $3 \nmid m$. As $\chi_3(2) = -1$, $\chi_3(m) = -\chi_3(j)s$, and inversion (Lemma 2.1(b)) gives $p = 2jQ - \chi_3(j)s$. Writing $j = 1 + 3i$ or $j = 2 + 3i$ yields $p \equiv 2Q - s$ or $4Q + s \pmod{6Q}$. Conversely let $p \equiv 2Q - s \pmod{6Q}$. Modulo 3, $p \equiv 2s - s = s$, so $\chi_3(p) = s$ and $m_0(p) = p + s \equiv (2Q - s) + s \equiv 0 \pmod{Q}$. If $p \equiv 4Q + s$, then modulo 3, $p \equiv s + s \equiv -s$, so $\chi_3(p) = -s$ and $m_0(p) = p - s \equiv 4Q \equiv 0 \pmod{Q}$. $\square$

Corollary 4.2 (Infinitely many owners). *Both classes in Lemma 4.1 are coprime to $6Q$: the numbers $2Q - s$ and $4Q + s$ are odd, are $\equiv s$ and $\equiv -s \pmod{3}$ respectively (so nonzero), and are $\equiv -s$ and $\equiv s \pmod{Q}$ respectively (so coprime to Q). By Dirichlet's theorem [2], each class contains infinitely many primes. Hence every odd Q coprime to 3 has infinitely many owners. In particular $\omega(Q) < \infty$ for every such Q , every odd prime $\neq 3$ is eventually a vertex, and by Corollary 3.3(b) $\omega(Q) \geq 2Q - 1$.*

The parametrisation (3) is also the search procedure: $\omega(Q)$ is the least prime among $p_1, p_2, p_4, p_5, p_7, \dots$. For example $\omega(M_{31}) = r_0 = 98\,784\,247\,763 = 46M_{31} + 1$ is $p_{23}(M_{31})$, since $\chi_3(23M_{31}) = -1$.

Theorem 4.3 (Bridging). *Let q_1, q_2 be odd primes different from 3. There is a window X such that q_1, q_2 lie in the same component of $G_{X'}$ for every $X' \geq X$.*

Proof. Put $Q = q_1q_2$ (or $Q = q_1$ if $q_1 = q_2$). Dirichlet step 1: by Corollary 4.2 choose a prime r with $Q \mid m_0(r)$; then $r \geq 5$ and r owns both q_i . Dirichlet step 2: choose a prime u with $r \mid m_0(u)$ and set $X = u$. Then $r \in S_X$. By door closure, $q_1, q_2 \in S_X$, and the arcs $r \rightarrow q_1$ and $r \rightarrow q_2$ are arcs of G_X ; so q_1, q_2 are joined through r . Persistence is (2). $\square$

Taking $q_1 = 5$ recovers the theorem of the hire-graph note that every odd prime other than 3 lies in the component of 5 in the infinite gold graph $G_\infty = \bigcup_X G_X$. (By Corollary 4.2, G_∞ has every odd prime $\neq 3$ as a vertex.)

Proposition 4.4 (Where the numbers come from). *Let q be an odd prime, $q \neq 3, 5$, and let r be a common owner of 5 and q with hirer u as in the proof of Theorem 4.3.*

- (a) (Lower bound, size bound.) $r \geq 10q - 1$ and $u \geq 2r - 1 \geq 20q - 3$. Hence no window below $20q - 3$ is joined to 5 by a bridge of this type.
- (b) (Upper bound, Linnik.) By Linnik's theorem with an admissible exponent L ($L = 5$ is known [3]), the least prime in a class modulo k is $\ll k^L$; so $r \ll q^L$ and $u \ll r^L \ll q^{L^2}$.

Proof. (a) $m_0(r)$ is an even multiple of $5q$, so $m_0(r) \geq 10q$ and $r \geq 10q - 1$; then $u \geq 2r - 1$ by Lemma 3.1. (b) is the two applications of Linnik's theorem, modulo $6 \cdot 5q$ and then modulo $6r$. $\square$

The window $[20q, q^{L^2}]$ is the whole unconditional gap between what the size bound forbids and what Linnik guarantees. Heuristically each of the two searches ends after a bounded multiple of log many attempts of size proportional to the modulus, consistent with the data for M_{31} (Section 7: $\omega(M_{31})/M_{31} = 46$ and $\omega(r_0)/r_0 = 14$), but that is a heuristic and is not used.

5 Isolation octaves

For a prime q , let

$$\alpha(q) = \min\{\omega(w) : w \text{ prime, } q \mid m_0(w)\}$$

be the first window in which some arc *into* q is live: an arc $w \rightarrow q$ needs an owner w of q that is itself a vertex, and $w \in S_X$ iff $\omega(w) \leq X$.

Theorem 5.1 (Isolation octave). *Let $M \in \mathcal{M}$ with $M \geq 7$.*

- (a) $\alpha(M) \geq 2\omega(M) - 1$.
- (b) For every X with $\omega(M) \leq X < \alpha(M)$, the prime M is an isolated vertex of G_X , and G_X is disconnected.
- (c) In particular gold is disconnected for all X with $\omega(M) \leq X \leq 2\omega(M) - 2$.

Proof. (a) If w owns M then $\omega(w) \geq 2w - 1 \geq 2\omega(M) - 1$ by Corollary 3.3(d). (b) $M \in S_X$ by (1). It has no outgoing arc, since $m_0(M)$ is a power of 2. An incoming arc $w \rightarrow M$ in G_X needs $M \mid m_0(w)$ and $w \in S_X$, that is $\omega(w) \leq X$, which contradicts $X < \alpha(M)$. Finally $\omega(M) \geq 2M - 1 \geq 13 > 11 = \omega(5)$, so $5 \in S_X$ is a second vertex, and $M \neq 5$. (c) follows from (a) and (b). $\square$

Compared with the first-owner window of the earlier version, this is a whole binary octave, and its length is at least $\omega(M) - 1 \geq 2M - 2$. Table 1 lists the intervals for every known 2-power door up to M_{31} ; for all of them below M_{31} , I verified by brute force that gold is disconnected on every window of the interval (`hire_checks.py`, part A).

The values 4111, 16381 and 1048573 are the satellites $\{257, 4111\}$, $\{8191, 16381\}$ and $\{524287, 1048573, \dots\}$ that appear in the window table of the hire-graph note; the theorem explains them as first owners, and shows they persist for a full octave.

M	$\omega(M)$	j	interval $[\omega, 2\omega - 2]$	checked
7	13	1	[13, 24]	yes
17	67	2	[67, 132]	yes
31	61	1	[61, 120]	yes
127	509	2	[509, 1 016]	yes
257	4 111	8	[4 111, 8 220]	yes
8 191	16 381	1	[16 381, 32 760]	yes
65 537	262 147	2	[262 147, 524 292]	yes
131 071	1 834 993	7	[1 834 993, 3 669 984]	yes
524 287	1 048 573	1	[1 048 573, 2 097 144]	yes
$2^{31} - 1$	98 784 247 763	23	[98 784 247 763, 197 568 495 524]	(Thm. 7.1)

Table 1: First owners and guaranteed isolation octaves; $m_0(\omega(M)) = 2jM$.

6 The equivalence

Theorem 6.1 (Connected stretches). *Let $11 \leq X_0 \leq X_1$. If G_{X_0} is connected and no $M \in \mathcal{M}$ has $X_0 < \omega(M) \leq X_1$, then G_X is connected for every $X_0 \leq X \leq X_1$.*

Proof. For such X we have $D_X = D_{X_0}$. The elements of D_{X_0} lie in one component of G_{X_0} , hence of G_X by (2). Apply Theorem 3.5(iii). $\square$

Theorem 6.2. *The following are equivalent.*

1. *There are infinitely many windows X on which undirected gold is disconnected.*
2. *There are infinitely many 2-power-door primes.*
3. *There are infinitely many Mersenne primes, or infinitely many Fermat primes.*

Proof. (2) $\Rightarrow$ (1). For each $M \in \mathcal{M}$ with $M \geq 7$, Theorem 5.1(c) gives disconnected windows starting at $\omega(M) \geq 2M - 1$, which tends to infinity with M .

(1) $\Rightarrow$ (2). Suppose $\mathcal{M}$ is finite. By Theorem 4.3, applied to the finitely many pairs from $\mathcal{M}$ and using (2), there is $X_0 \geq \max_{M \in \mathcal{M}} \omega(M)$ such that all elements of $\mathcal{M}$ lie in one component of G_{X_0} . For every $X \geq X_0$ we have $D_X = \mathcal{M}$, and these primes remain in one component by (2). By Theorem 3.5(iii), G_X is connected for every $X \geq X_0$, so only finitely many windows are disconnected.

(2) $\Leftrightarrow$ (3) is the classification of 2-power doors as Mersenne primes with odd exponent and Fermat primes greater than 3. $\square$

Fermat primes are believed finite (five are known, the largest 65537), and the infinitude of Mersenne primes is open; so Theorem 6.2 identifies the obstruction rather than settling item (1). Two features are new relative to the earlier proof of (1) $\Rightarrow$ (2): the argument no longer speaks of a sink being “absorbed” informally, but uses the exact criterion of Theorem 3.5; and the direction (2) $\Rightarrow$ (1) produces intervals, so a disconnected window is never an isolated event.

Remark 6.3 (Heuristic). The Lenstra–Pomerance–Wagstaff heuristic [4] predicts about $e^\gamma \log_2 x$ Mersenne prime exponents up to x , hence infinitely many, and by Theorem 6.2 infinitely many disconnected windows. Near $X \approx 2^p$ an octave is disconnected only if p is a Mersenne exponent, which is expected with probability of order $e^\gamma / (p \log 2)$; if the reconnection lengths $\alpha(M)/\omega(M)$ are typically bounded, the disconnected windows would have logarithmic density zero yet be infinite in number. This is a labelled heuristic; the bound on $\alpha(M)/\omega(M)$ is not proved, and nothing here depends on it.

7 M_{31} , exactly

Let $M = M_{31} = 2^{31} - 1 \equiv 1 \pmod{3}$, so $s = +1$. By Lemma 4.1 the owners of M are the primes $tM - 1$ with $t \equiv 2 \pmod{6}$ and $tM + 1$ with $t \equiv 4 \pmod{6}$ ($t = 2j$). The least is

$$r_0 = \omega(M_{31}) = 46M + 1 = 98\,784\,247\,763, \quad m_0(r_0) = 2 \cdot 23 \cdot M_{31},$$

and no smaller t gives a prime; the floor candidate $2M - 1 = 4\,294\,967\,293 = 9241 \cdot 464773$ is composite. Two searches of the same kind give the hire time of r_0 itself and all owners that could matter. The hire time of r_0 is

$$\omega(r_0) = 14r_0 + 1 = 1\,382\,979\,468\,683,$$

the least prime among the numbers $tr_0 \mp 1$ with t even and $\equiv 2, 4 \pmod{6}$ respectively. An owner w of M_{31} can be hired only at $X \geq 2w - 1$ (Corollary 3.3(b)), so the only owners that can be hired before $\omega(r_0)$ satisfy $2w - 1 \leq 1\,382\,979\,468\,683$. There are exactly thirteen:

t	owner $w = tM_{31} \mp 1$	door $m_0(w) = tM_{31}$	hire time $\omega(w)$
46	98 784 247 763	$2 \cdot 23 \cdot M_{31}$	1 382 979 468 683
136	292 057 775 993	$2^3 \cdot 17 \cdot M_{31}$	$> 1.383 \cdot 10^{12}$
140	300 647 710 579	$2^2 \cdot 5 \cdot 7 \cdot M_{31}$	$> 1.383 \cdot 10^{12}$
146	313 532 612 461	$2 \cdot 73 \cdot M_{31}$	627 065 224 921
184	395 136 991 049	$2^3 \cdot 23 \cdot M_{31}$	$> 1.383 \cdot 10^{12}$
196	420 906 794 813	$2^2 \cdot 7^2 \cdot M_{31}$	841 813 589 627
220	472 446 402 341	$2^2 \cdot 5 \cdot 11 \cdot M_{31}$	$> 1.383 \cdot 10^{12}$
226	485 331 304 223	$2 \cdot 113 \cdot M_{31}$	$> 1.383 \cdot 10^{12}$
238	511 101 107 987	$2 \cdot 7 \cdot 17 \cdot M_{31}$	$> 1.383 \cdot 10^{12}$
242	519 691 042 573	$2 \cdot 11^2 \cdot M_{31}$	$> 1.383 \cdot 10^{12}$
272	584 115 551 983	$2^4 \cdot 17 \cdot M_{31}$	$> 1.383 \cdot 10^{12}$
280	601 295 421 161	$2^3 \cdot 5 \cdot 7 \cdot M_{31}$	$> 1.383 \cdot 10^{12}$
296	635 655 159 511	$2^3 \cdot 37 \cdot M_{31}$	$> 1.383 \cdot 10^{12}$

Table 2: Owners w of M_{31} with $2w - 1 \leq \omega(r_0)$. The entry $> 1.383 \cdot 10^{12}$ means no prime owner of w exists up to $\omega(r_0)$.

The minimum is attained by $w = 313\,532\,612\,461 = 146M_{31} - 1$, and

$$X_1 := \alpha(M_{31}) = 627\,065\,224\,921 = 2w - 1,$$

the extremal case of the size bound (Remark 3.2): w is hired by the prime $2w - 1$, whose door is $2w$. The door of w is $2 \cdot 73 \cdot M_{31}$, so at X_1 there are arcs $w \rightarrow M_{31}$ and $w \rightarrow 73$, and 73 lies in the mainland by the chain

$$73 \rightarrow 37 \rightarrow 19 \rightarrow 5, \quad m_0(73) = 2 \cdot 37, \quad m_0(37) = 2 \cdot 19, \quad m_0(19) = 2^2 \cdot 5.$$

Theorem 7.1 (The M_{31} stretch). *Assume that G_{X^*} is connected for $X^* = 92\,274\,421$ (hire-graph note, machine-checked, not Lean). Then for every X with $X^* \leq X \leq 2M_{61} - 2 = 2^{62} - 4$, the gold graph G_X is disconnected if and only if*

$$98\,784\,247\,763 = \omega(M_{31}) \leq X \leq \alpha(M_{31}) - 1 = 627\,065\,224\,920.$$

Proof. Which sinks occur. If $M \in \mathcal{M}$ is in S_X then $M \leq (X + 1)/2 < M_{61}$ when $X \leq 2M_{61} - 2$. The 2-power doors below M_{61} are the Mersenne primes with exponent ≤ 31 (odd exponents 3, 5, 7, 13, 17, 19, 31) and the Fermat primes 5, 17, 257, 65537 ($F_5 = 641 \cdot 6700417$ is composite and $F_6 > M_{61}$). So $\mathcal{M} \cap [1, M_{61}) = \{5, 7, 17, 31, 127, 257, 8191, 65537, 131071, 524287, M_{31}\}$, and

all except M_{31} have $\omega \leq 1\,834\,993 < X^*$ (Table 1). Hence $D_X = D_{X^*}$ for $X^* \leq X < \omega(M_{31})$, and $D_X = D_{X^*} \cup \{M_{31}\}$ for $\omega(M_{31}) \leq X \leq 2M_{61} - 2$.

Before $\omega(M_{31})$. D_{X^*} lies in one component of G_{X^*} (connected), hence of G_X by (2); Theorem 6.1 gives connectedness on $[X^*, \omega(M_{31}) - 1]$.

On $[\omega(M_{31}), X_1 - 1]$. By Theorem 5.1(b), M_{31} is isolated and gold is disconnected, provided $\alpha(M_{31}) = X_1$. That holds because every owner w of M_{31} with $\omega(w) < X_1$ would satisfy $2w - 1 \leq \omega(w) < X_1 < \omega(r_0)$, so it appears in Table 2; and there the smallest hire time is X_1 .

From X_1 on. At X_1 the vertex w is present with arcs to M_{31} and to 73; by door closure 37, 19, 5 $\in S_{X_1}$ and the chain above is live, so M_{31} is joined to 5, hence to all of D_{X^*} . Thus D_{X_1} is in one component, and by (2) it stays so on $[X_1, 2M_{61} - 2]$, where no further sink enters; Theorem 3.5 gives connectedness. $\square$

Remark 7.2 (The old bridge owner). The previous version certified $v = 300\,647\,710\,579 = 140M_{31} - 1$, with $m_0(v) = 2^2 \cdot 5 \cdot 7 \cdot M_{31}$, as the least common owner of 5 and M_{31} and noted that a bridge *owner* is not a bridge *vertex*. Table 2 settles this: v has no owner up to $1.383 \cdot 10^{12}$, far later than the actual reconnection $X_1 \approx 6.27 \cdot 10^{11}$. The reconnection occurs through a different owner, one whose door has no factor 5; M_{31} reaches 5 only through $73 \rightarrow 37 \rightarrow 19 \rightarrow 5$. It also confirms that the “first direct 5– M_{31} bridge” is not the first connecting vertex.

Remark 7.3 (Cost). The earlier version called a full component check at these scales “past home computing”. The exact answer above needs only Corollary 4.2’s parametrisation (about 650 candidates $tM_{31} \mp 1$ with $t \leq 322$) and a few dozen primality tests of numbers below $2 \cdot 10^{12}$; see `hire_checks.py`, part B. The reason is Theorem 3.5: connectivity is decided by the sinks, so no graph rebuild is required.

8 Formalisation, reproducibility and changes

Lean. The Lean development (`Hire/GoldDisconnects.lean`) previously checked the single-window isolation lemma, the descent lemma, and the implication from infinitely many 2-power doors to infinitely many disconnected windows; door closure was not checked, so the converse half of Theorem 6.2 waited on it. The results new in this revision (Theorem 3.5, Theorem 5.1 in its interval form, Theorem 6.1, Theorem 7.1) are not formalised. The computed data (Tables 1 and 2, the value X_1) are certificates, not Lean theorems.

Reproducibility. `hire_checks.py` (Python 3, `sympy`) performs two tasks. Part A replays all windows $X \leq L$ (default 10^6 ; 10^7 was used for the statements above) and checks Theorem 3.5(iii) at each window and disconnectedness on each interval of Table 1 up to L . Part B recomputes r_0 , $\omega(r_0)$, the thirteen owners, their hire times, X_1 , and the chain $73 \rightarrow 37 \rightarrow 19 \rightarrow 5$. Primality of numbers below 2^{64} is decided by `sympy.isprime`, which is deterministic in that range.

Changes from the earlier version. (1) Lemma 3.1 now records $q \leq (p+1)/2$; the previous “ $< (p+1)/2$ ” was false for $m_0(p) = p+1$. (2) The single first-owner window became an octave (Theorem 5.1). (3) The informal “absorbed into the infinite component” argument became Theorem 3.5 and Theorem 6.1. (4) The M_{31} data became an exact statement about all X up to $2M_{61} - 2$, and the bridge owner v is shown not to be the bridge. Everything else in the earlier note is retained; the statement of the equivalence is unchanged.

References

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